

## Eigenvalues

## Overview

- geometric understanding for eigenvalues and eigenvectors and how they relate to matrices
- matrix transformation using diagonalization
- special properties of symmetric matrices
- orthogonal matrices, rotations and projection
- SPD matrices and connection to quadratic forms
- convolution, DFTs and eigen-decomposition

## Async

Geometric Interpretation of  $Ax$ 

## Mapping

- eigenvalue **decomposition** can reveal **structure** of a matrix as an **operator**

$$A = S\Lambda S^{-1}$$

- **apply** to  $\vec{x}$  and move **right to left**

$$Ax = S(\Lambda(S^{-1}x))$$

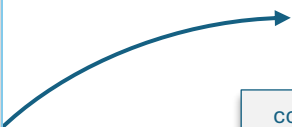
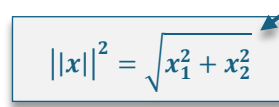
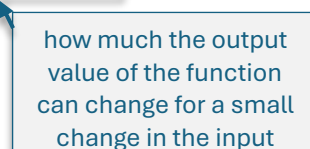
- example:

$$A = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} \quad S = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$

## Julia

```
julia> A = [3 -1; -1 3]
2×2 Matrix{Int64}:
 3  -1
-1   3

julia> eigen(A)
EigenFloat64{Float64, Matrix{Float64}, Vector{Float64}}
values:
2-element Vector{Float64}:
 2.0
 4.0
vectors:
2×2 Matrix{Float64}:
 0.707107 -0.707107
 0.707107  0.707107
```

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| <b>1. Coordinate</b>       | $Ax = S(\Lambda(S^{-1}x))$ <ul style="list-style-type: none"> <li>definition of an <b>inverse</b> 2 x 2 <b>matrix</b> <math display="block">S^{-1} \begin{bmatrix} a &amp; b \\ c &amp; d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d &amp; -b \\ -c &amp; a \end{bmatrix}</math> </li> <li>so, for the <b>matrix</b> <math display="block">S = \begin{bmatrix} 1 &amp; -1 \\ 1 &amp; 1 \end{bmatrix}, \quad S^{-1} = \frac{1}{2} \begin{bmatrix} 1 &amp; 1 \\ -1 &amp; 1 \end{bmatrix}</math> <div style="display: flex; justify-content: space-around; margin-top: 10px;"> <div style="text-align: center;">  <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 0 auto;">counterclockwise 45°<br/>scale up by <math>\sqrt{2}</math></div> </div> <div style="text-align: center;">  <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 0 auto;">clockwise 45°<br/>scale down by <math>\sqrt{2}</math></div> </div> </div> </li> </ul> |
| <b>2. Scale</b>            | <ul style="list-style-type: none"> <li>multiplying by the <b>eigenvalues</b> in matrix <math>\Lambda</math> <b>scales</b> each vector <b>independently</b> <math display="block">\Lambda = \begin{bmatrix} 2 &amp; 0 \\ 0 &amp; 4 \end{bmatrix}</math> <div style="display: flex; justify-content: flex-end; margin-top: 10px;"> <div style="text-align: center; margin-right: 20px;">  <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 0 auto;">scales x by 2</div> </div> <div style="text-align: center;">  <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 0 auto;">scales y by 4</div> </div> </div> </li> </ul>                                                                                                                                                                                                                                                                                                                          |
| <b>3. Map Back</b>         | <ul style="list-style-type: none"> <li>finally, <b>map</b> back with <math>S</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Putting it Together</b> | $Ax = S(\Lambda(S^{-1}x))$ $Ax = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \left( \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} x \right) = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} x$ $Ax = \begin{bmatrix} 3x_1 - x_2 \\ -x_1 + 3x_2 \end{bmatrix}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Symmetric Matrices</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Properties</b>          | <ul style="list-style-type: none"> <li>have <b>real</b> eigen <b>values</b> and <b>vectors</b></li> <li><b>orthogonal</b> eigenvectors that <b>span</b> the <b>space</b></li> <li>can be <b>broken down</b> into rank 1 matrices</li> <li>easy to <b>compute</b> the 2-D norm and <b>condition number</b></li> </ul> <div style="display: flex; justify-content: space-around; margin-top: 20px;"> <div style="text-align: center;">  <div style="border: 1px solid black; padding: 10px; width: fit-content; margin: 0 auto;"><math>\ x\ ^2 = \sqrt{x_1^2 + x_2^2}</math></div> </div> <div style="text-align: center;">  <div style="border: 1px solid black; padding: 10px; width: fit-content; margin: 0 auto;">how much the output value of the function can change for a small change in the input</div> </div> </div>                                                                                                                                                                              |

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| <b>Why Real Eigenvalues</b> | <ul style="list-style-type: none"> <li>a <b>matrix</b> always has at least <b>one eigenvalue</b> (possibly <b>complex</b>)<br/> <math display="block">p(\lambda) = \det(A - \lambda I) = (-1)^n \lambda^n + \dots + \det(A)</math></li> <li>if, <math>Au = \lambda u</math></li> <li>for matrices, the <b>dot product</b> uses the <b>complex conjugate</b> transpose<br/> <math display="block">(Au)^H u = (\lambda u)^H u = \bar{\lambda} u^H u = \bar{\lambda} \ u\ ^2</math> <math display="block">(Au)^H u = u^H A^H u = u^H A u = u^H (\lambda u) = \lambda u^H u = \lambda \ u\ ^2</math></li> <li>which <b>implies</b>,<br/> <math display="block">\bar{\lambda} = \lambda \quad \lambda \in \mathbb{R}</math> <div data-bbox="1112 630 1446 772" style="border: 1px solid black; padding: 5px; margin-top: 10px;"> <math>\bar{z}</math> : obtained by <b>reversing</b> the <b>sign</b> of the <b>imaginary</b> part while <b>keeping</b> the <b>real</b> part the <b>same</b> </div> </li> </ul>                               |
| <b>Why Orthogonal</b>       | <ul style="list-style-type: none"> <li>a key benefit of <b>symmetric matrices</b> is that the <b>eigenvectors</b> are <b>orthogonal</b><br/> <math display="block">A u_1 = \lambda_1 u_1, \quad A u_2 = \lambda_2 u_2, \quad \lambda_1 \neq \lambda_2 \Rightarrow u_1 \cdot u_2 = 0</math></li> <li>why,<br/> <math display="block">u_1^T A u_2 = u_1^T (A u_2) = u_1^T (\lambda_2 u_2) = \lambda_2 u_1^T u_2 = \lambda_2 (u_1 \cdot u_2)</math> <math display="block">u_1^T A u_2 = (A^T u_1)^T u_2 = (A u_1)^T u_2 = \lambda_1 u_1^T u_2 = \lambda_1 (u_1 \cdot u_2)</math></li> <li>since the <b>eigenvalues</b> are <b>different</b>, the <b>dot product</b> has to be <b>zero</b><br/> <math display="block">\lambda_1 (u_1 \cdot u_2) = \lambda_2 (u_1 \cdot u_2)</math> <math display="block">(\lambda_1 - \lambda_2) (u_1 \cdot u_2) = 0</math> <math display="block">u_1 \cdot u_2 = 0</math></li> <li><b>normalize</b> the <b>eigenvectors</b> so,<br/> <math display="block">\ u_i\ ^2 = u_i \cdot u_i = 1</math></li> </ul> |

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| <b>Orthogonal<br/>Diagonalization</b> | <ul style="list-style-type: none"> <li>another benefit of <b>symmetric matrices</b> is that you <b>have</b> <b><math>n</math></b> orthogonal eigenvectors <math display="block">q_i = \frac{u_i}{  u_i  }</math> </li> <li><b>stack</b> them into a <b>matrix</b> <math display="block">Q = [q_1; q_2; \dots; q_n]</math> </li> <li>this is an <b>orthogonal square</b> matrix <math display="block">Q^T Q = I \Rightarrow Q^{-1} = Q^T</math> </li> <li>this <b>avoids</b> explicitly <b>computing</b> a matrix <b>inverse</b>, which is <b>required</b> for <b>general</b> (non-symmetric) <b>matrices</b></li> </ul>                                                 |
| <b>Spectral<br/>Decomposition</b>     | <ul style="list-style-type: none"> <li>this gives you a <b>real valued factorization</b> <math display="block">A = Q \Lambda Q^T</math> <math display="block">A = Q \Lambda Q^T = [q_1; q_2; \dots; q_n] \begin{bmatrix} \lambda_1 q_1^T \\ \cdot \\ \cdot \\ \cdot \\ \lambda_n q_n^T \end{bmatrix}</math> </li> <li>apply this to a <b>point</b> <math>x</math> <math display="block">Ax = \lambda_1 q_1 q_1^T x + \lambda_2 q_2 q_2^T x + \dots + \lambda_n q_n q_n^T x</math> </li> <li><b>equivalent scalar</b> form <math display="block">Ax = \lambda_1 (q_1 \cdot x) q_1 + \lambda_2 (q_2 \cdot x) q_2 + \dots + \lambda_n (q_n \cdot x) q_n</math> </li> </ul> |
| <b>Inverse</b>                        | <ul style="list-style-type: none"> <li>if <b>none</b> of the eigenvalues are <b>zero</b>, <b><math>A</math></b> is <b>invertible</b> <math display="block">A^{-1} = Q \Lambda^{-1} Q^T \quad A = Q \Lambda Q^T</math> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Quadratic Forms – SPD              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Symmetric Positive Definite</b> | <ul style="list-style-type: none"> <li>symmetric matrices are always diagonalizable and have <b>real eigenvalues</b></li> <li>if the eigenvalues are <b>all positive</b>, the matrix is called <b>Symmetric Positive Definite (SPD)</b></li> <li>if they are <b>all negative</b>, the matrix is <b>symmetric negative definite</b></li> <li>if they are <b>all non-negative</b> (<math>\geq 0</math>), the matrix is <b>symmetric positive semidefinite</b></li> <li><b>projection</b> expression <math display="block">q_k q_k^T x = (q_k \cdot x) q_k</math> </li> <li>spectral <b>expansion</b> applied to a <b>vector</b> <math display="block">Ax = \lambda_1 q_1 q_1^T x + \lambda_2 q_2 q_2^T x + \cdots + \lambda_n q_n q_n^T x</math> </li> <li><b>equivalent</b> projection form <math display="block">Ax = \lambda_1 (q_1 \cdot x) q_1 + \lambda_2 (q_2 \cdot x) q_2 + \cdots + \lambda_n (q_n \cdot x) q_n</math> </li> </ul> |
| <b>Quadratic Forms</b>             | <ul style="list-style-type: none"> <li>example of a <b>quadratic</b> form <math display="block">f(x, y) = 5x^2 - 2xy + 5y^2</math> </li> <li><b>matrix</b> form <math display="block">  \begin{aligned}  f(x, y) &amp;= 5x^2 - xy - yx + 5y^2 \\  &amp;= x(5x - y) + y(-x + 5y) \\  &amp;= [x \ y] \begin{bmatrix} 5 &amp; -1 \\ -1 &amp; 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\  &amp;= [x \ y] \begin{bmatrix} 5x - y \\ -x + 5y \end{bmatrix} \\  &amp;= \begin{bmatrix} x \\ y \end{bmatrix}^T Q \Lambda Q^T \begin{bmatrix} x \\ y \end{bmatrix} \\  &amp;= u^T \Lambda u  \end{aligned}  </math> </li> </ul>                                                                                                                                                                                                                                                                                                       |

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| <b>Julia</b>       | <pre> julia&gt; A = [5 -1; -1 5] 2×2 Matrix{Int64}:  5  -1 -1   5  julia&gt; eigen(A) Eigen{Float64, Float64, Matrix{Float64}, Vector{Float64}} values: 2-element Vector{Float64}:  4.0  6.0 vectors: 2×2 Matrix{Float64}: -0.707107 -0.707107 -0.707107  0.707107 </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Summary</b>     | <ul style="list-style-type: none"> <li>In general, for all <b>symmetric</b> matrices <math display="block">x^T A x = (Q^T x)^T \Lambda (Q^T x) = u^T \Lambda u = \lambda_1 u_1^2 + \dots + \lambda_n u_n^2</math> </li> </ul> <p>If the <b>matrix</b> is:</p> <ul style="list-style-type: none"> <li><b>SPD</b> <ul style="list-style-type: none"> <li>then all <b>eigenvalues</b> are strictly <b>greater</b> than 0</li> <li>the form is <b>convex</b></li> <li>passes <b>through</b> the <b>origin</b></li> </ul> </li> <li><b>negative</b> definite <ul style="list-style-type: none"> <li>0 is the <b>unique maximum</b></li> </ul> </li> <li><b>some</b> eigenvalues are <b>negative</b> and others <b>positive</b> <ul style="list-style-type: none"> <li>the <b>quadratic</b> form has a <b>saddle point</b></li> </ul> </li> </ul> |
| <b>Matrix Norm</b> | <ul style="list-style-type: none"> <li>for matrix <b>norms</b>, we define <math display="block">\ A\  = \max_{x \neq 0} \frac{\ Ax\ }{\ x\ } = \max_{\ x\ =1} \ Ax\ </math> </li> <li><b>Euclidean distance</b> formula (<b>2-norm</b>) <math display="block">\ Ax\ ^2 = (Ax)^T (Ax) = x^T A^T A x = (Q^T x)^T \Lambda (Q^T x)</math> </li> <li>change of variables <math display="block">\ x\  = 1, \quad u = Q^T x \Rightarrow \ u\  = \ x\  = 1</math> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                       |

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|                                        | <ul style="list-style-type: none"> <li>• bounding <b>expressions</b><br/> <math display="block">  Ax  ^2 = \lambda_1 u_1^2 + \dots + \lambda_n u_n^2 \leq \lambda_{\max}(u_1^2 + \dots + u_n^2) = \lambda_{\max}</math> </li> <li>• <b>matrix 2-norm</b><br/> <math display="block"> A _2 = \sqrt{(\lambda_{\max}(A^T A))} = \sigma_{\max}(A)</math> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Eigenvalues for 1-D Convolution</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Convolution: 1D</b>                 | <ul style="list-style-type: none"> <li>• the <b>DFT</b> links <b>convolutions</b> and eigen <b>decompositions</b></li> <li>• simple 1-D <b>convolution</b>: <ul style="list-style-type: none"> <li>○ input: <math>\vec{u}</math></li> <li>○ output: <math>\vec{v}</math></li> </ul> <math display="block">v_j = \frac{u_{j-1}}{4} + \frac{u_j}{2} + \frac{u_{j+1}}{4}</math> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Convolution: Eigenvector</b>        | <ul style="list-style-type: none"> <li>• use <b>vectors</b> from <b>DFT</b><br/> <math display="block">u_j = \exp\left(\frac{i2\pi kj}{n}\right)</math> </li> <li>• <b>evaluate</b><br/> <math display="block">v_j = \frac{u_{j-1}}{4} + \frac{u_j}{2} + \frac{u_{j+1}}{4}</math> <math display="block">= \frac{1}{4} \exp\left(\frac{i2\pi k(j-1)}{n}\right) + \frac{1}{2} \exp\left(\frac{i2\pi kj}{n}\right) + \frac{1}{4} \exp\left(\frac{i2\pi k(j+1)}{n}\right)</math> <math display="block">= \exp\left(\frac{i2\pi kj}{n}\right) \left( \frac{1}{4} \exp\left(-\frac{i2\pi k}{n}\right) + \frac{1}{2} + \frac{1}{4} \exp\left(\frac{i2\pi k}{n}\right) \right)</math> <math display="block">= \underbrace{\exp\left(\frac{i2\pi kj}{n}\right)}_{u_j} \underbrace{\left( \frac{1}{2} \cos\left(\frac{2\pi k}{n}\right) + \frac{1}{2} \right)}_{\text{constant}}</math> </li> </ul> |

## Convolution Operator

- consider,

$$v_j = \frac{u_{j-1}}{4} + \frac{u_j}{2} + \frac{u_{j+1}}{4}$$

- with periodic **boundary condition**

$$v = L(u), L: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

- the **Fourier** frequency vector

$$q_k = \left( \frac{1}{\sqrt{n}} \exp\left(\frac{i2\pi kj}{n}\right) \right)_{j=1, \dots, n}$$

- eigenvector** relation

$$L(q_k) = \lambda_k q_k$$

- eigenvalue**

$$\lambda_k = \frac{\cos\left(\frac{2\pi k}{n}\right) + 1}{2}$$

- diagonalization** of the operator

$$L(u) = Q \Lambda Q^H u$$



### Convolution: DFT

- **since the Fourier frequencies are eigenvectors for the convolution operator, we can use that to evaluate the convolution**

- **applying the operator (in a periodic domain)**

$$v_j = \frac{u_{j-1}}{4} + \frac{u_j}{2} + \frac{u_{j+1}}{4}$$

- **equivalent matrix form**

$$v = Q\Lambda Q^H u$$

- **eigenvalues**

$$\lambda_k = \frac{\left(\cos\left(\frac{2\pi k}{n}\right) + 1\right)}{2}$$

- **using the DFT:**

$$Q^H u \quad \text{apply the Fourier transform to } u$$

$$\Lambda Q^H u \quad \text{scale each } k \text{ component by the eigenvalue for the operator}$$

$$Q\Lambda Q^H u \quad \text{apply the inverse Fourier transform to the output}$$